

Connecting to Database in JSP

- All the web applications usually interact with different types of databases. Databases are basically used to store various types of information for different purposes.
- JSP pages are looks like regular html pages with the inclusion of dynamic behavior.
- JSP pages allows the developer to interact with database by using JDBC code.
- By using Scriptlets, we can include database programming (JDBC code) in JSP.

- The Architecture of accessing a database by using JSP.

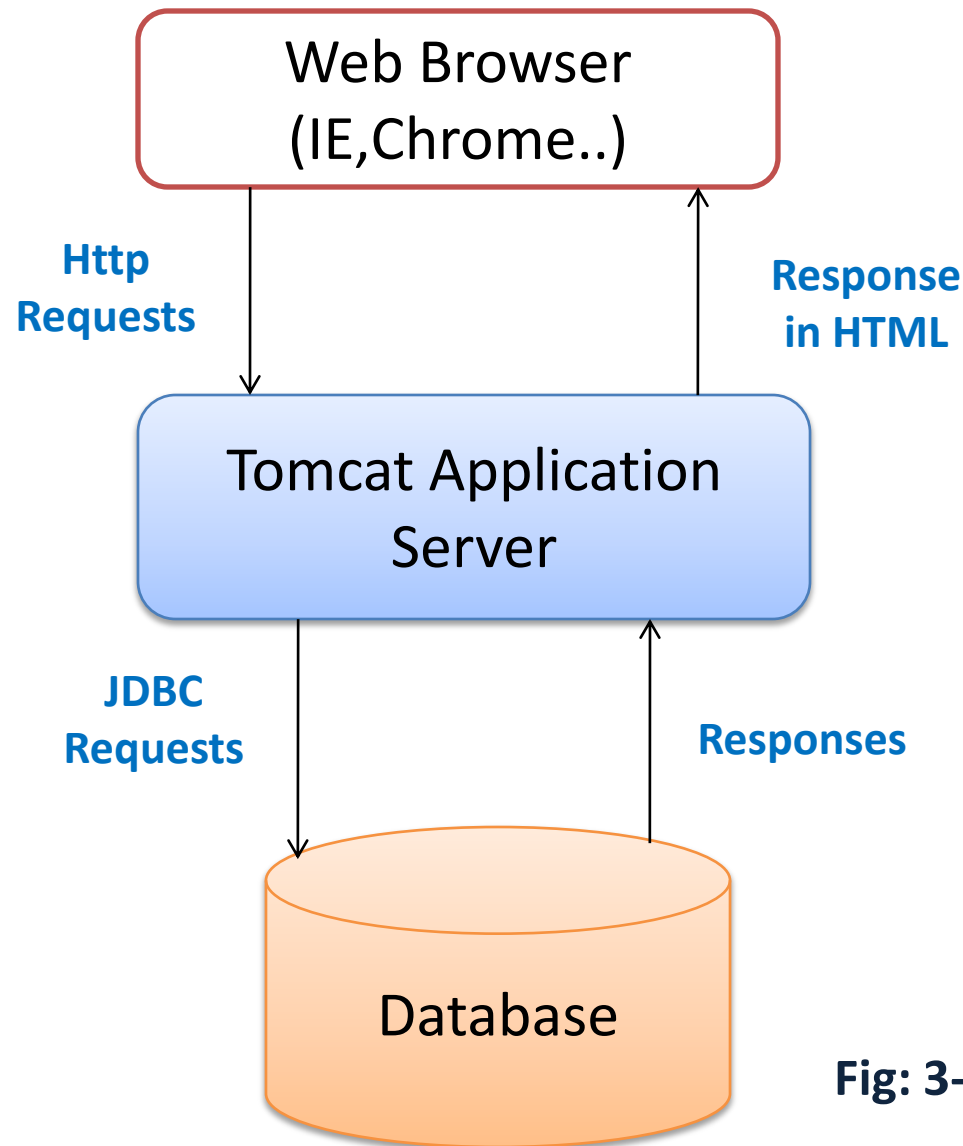


Fig: 3-tier Architecture

- For Database Connectivity we write the following code in jsp page.

✓ **Open Connectivity Code:**

```
<%@page language="java" import="java.sql.*"%>
```

```
<%
```

```
// Load driver class file of Oracle
```

```
class.forName("oracle.jdbc.driver.OracleDriver");
```

```
Connection con=DriverManager.getConnection(
```

```
"jdbc:oracle:thin:@localhost:1521:xe","system","password");
```

```
// Load driver class file of MySql
```

```
class.forName("com.mysql.jdbc.Driver");
```

```
Connection con=DriverManager.getConnection(
```

```
" jdbc:mysql://localhost:3306/TestDb","root","root");
```

```
%>
```

✓ **Statement Code:**

<%

// Create the statement

Statement stmt=conn.createStatement();

String query="SELECT * from students"

stmt.executeQuery(query);

%>

✓ **Close connectivity Code:**

<%

// Close the Statement

stmt.close();

// Close the connection

conn.close();

%>

Example: Registration.html

```
<html >
  <head> <title>Registration</title> </head>
  <body>
    <form action="reg.jsp" method="post">
      Email :<input type="text" name="email" /><br/>
      First name :<input type="text" name="fname" /><br/>
      Last name :<input type="text" name="lname" /><br/>
      User name :<input type="text" name="userid" /><br/>
      password :<input type="password" name="pwd" /><br/>
      <input type="submit" />
    </form>
  </body>
</html>
```

reg.jsp

```
<%@ page import = "java.sql.*" %>
```

```
<%@ page import = "javax.sql.*" %>
```

```
<%
```

```
String user=request.getParameter("userid");
```

```
String pwd=request.getParameter("pwd");
```

```
String fname=request.getParameter("fname");
```

```
String lname=request.getParameter("lname");
```

```
String email=request.getParameter("email");
```

```
Class.forName("oracle.jdbc.driver.OracleDriver");
```

```
Connection con = DriverManager.getConnection  
    ("jdbc:oracle:thin:@localhost:1521:xe","system","system");
```

```
Statement st= con.createStatement();
```

```
String query="insert into users values
```

```
    ('"+user+"','"+pwd+"','"+fname+"','"+lname+"','"+email+"')";
```

```
st.executeUpdate(query);
```

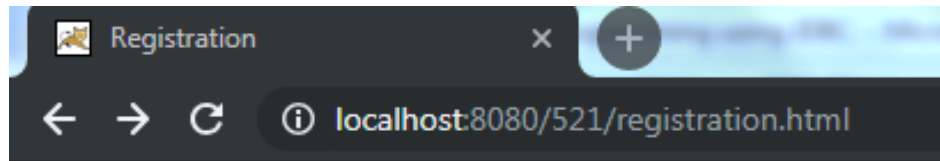
```
st.close();
```

```
con.close();
```

```
out.println("registration successfull");
```

```
%>
```

Output:



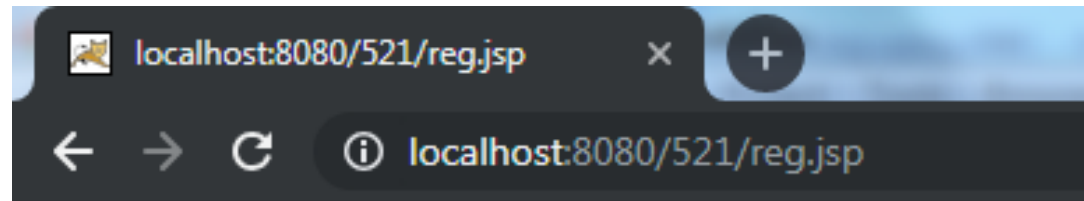
Email :

First name :

Last name :

User name :

password :



registration successfull